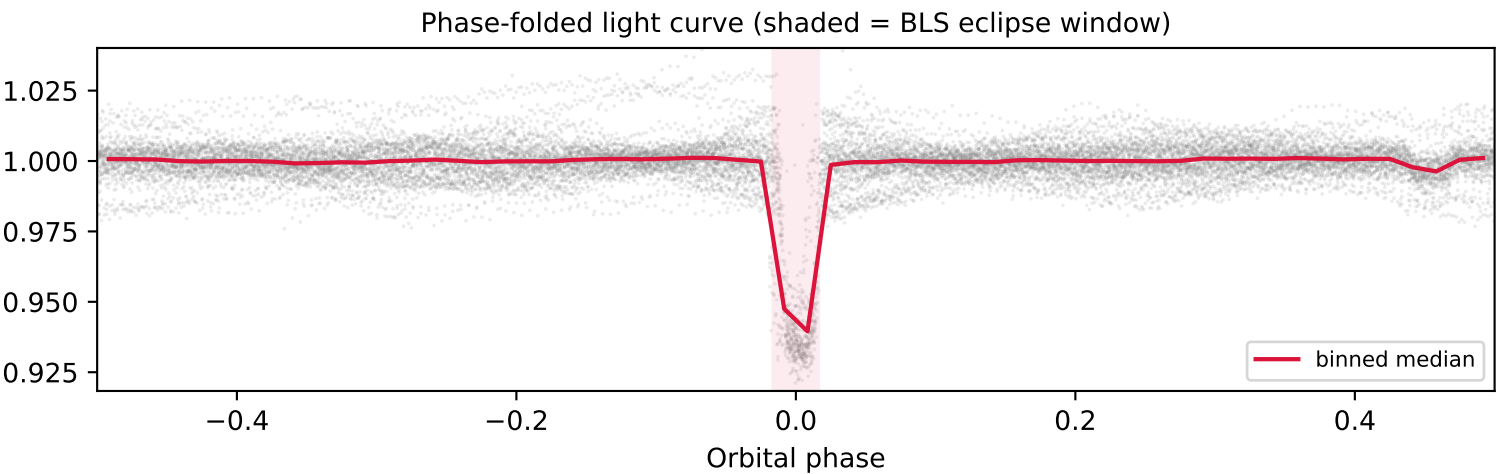


TIC 220423668

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	3.651055 d (ALIASED, see below)	VSX	no match
-> corrected period:	7.302111 d	ASAS-SN	no match
Epoch (t0)	1413.9498 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	4.525 % (45,247 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	755.4	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	55 (gate >= 3)		
Robust depth ratio	1.27 (gate >= 0.50)		
Robust depth sigma	215.5 (gate >= 7.0)		
Odd/even z-score	8.94 (ALIAS FLAGGED)		
Secondary eclipse	not detected		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 32	light-curve dip in this sector: 83.6 sigma	t0 refined by 0.0025 phase	
Difference-image peak SNR	17.67 (gate >= 5.0)	Compactness	0.75 (gate >= 0.50)
Dimming centroid offset	0.28 px (6 arcsec) (gate <= 1.0 px, i.e. on the target)		
Nearest bright-enough neighbour in cutout: TIC 220423674			
	3.23 px away, dTmag +1.58 - farther from the dimming than the target		



difference-image figure not found

Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 8.94 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 7.302111 d, in any write-up.